Level 2 — Generator Forensics Report

Project

Hackathon: AI & Data Challenge — Intelligent Candidate Ranking System

Phase: Level 2 — Dataset Generator Reverse Engineering

Objective:

Reverse engineer the synthetic candidate generator to understand:

- How candidate profiles were created
- Which fields are trustworthy
- Which fields are noisy
- Hidden latent variables
- Relationships between profile components
- Potential ranking features for the final system

Executive Summary

This investigation analyzed 100,000 candidate profiles to identify:

- Template generation patterns
- Skill generation logic
- Candidate archetypes
- Behavioral signal generation
- Hidden candidate quality factors
- Missing value encoding strategies

The goal was not merely exploratory analysis, but to understand how the dataset itself was constructed and which signals should be trusted when building the ranking engine.

Section 2.1 — Summary Template Discovery

Objective

Determine whether profile summaries were written uniquely or generated from templates.

Methodology

A deep analysis was performed to identify and evaluate summary template patterns across the dataset. Candidate summaries were extracted, cleaned, and normalized by removing variable elements such as years of experience, specific technologies, company names, and other dynamic content. This process enabled the detection of underlying template structures rather than surface-level variations.

The analysis revealed a total of 3,241 unique summary templates before normalization. After normalization, these were reduced to 74 unique normalized templates, indicating extensive reuse of common summary structures throughout the dataset. This normalization-based approach provided strong evidence that summaries were generated from a relatively small set of reusable templates.

Describe:

- How summaries were extracted
- How normalization was performed
- How templates were identified

Evidence

Evidence File:

```
evidence/summary_templates.csv
```

Evidence Screenshot:

```
[UPLOAD SCREENSHOT]
```

Findings

Total Summary Templates

```
76
```

Observation

Large portions of candidate summaries reuse nearly identical sentence structures.

Example Pattern:

```
Professional with X years of experience...
Open to AI opportunities...
Interested in applying domain expertise...
```

Meaning

Summarys appear heavily template-generated.

Candidate individuality is weak inside summaries.

Trustworthiness

Component	Trust Score
Summary Text	LOW

Impact On Ranking

Do not rely heavily on summaries.

Summaries should contribute weak contextual evidence only.

Section 2.2 — Description Template Discovery

Objective

Determine whether work-history descriptions were uniquely authored.

Methodology

Work history provides contextual evidence about a candidate’s career progression, domain expertise, and practical experience. While individual job descriptions may contain synthetic elements, patterns across roles, industries, tenure, and responsibilities can still help validate skill claims, identify relevant experience, and improve candidate-job matching. As a result, work history should be treated as a supporting signal that complements skills and behavioral indicators rather than serving as a standalone ranking factor.

Evidence

Evidence File:

```
evidence/description_templates.csv
```

Evidence Screenshot:

```
[UPLOAD]
```

Findings

Total Description Templates

```
44
```

Observation

Many candidates share identical job descriptions despite different:

- Titles
- Industries
- Companies

Example:

Marketing Manager

↓

Mechanical Engineering Description

Business Analyst

↓

Accounting Description

Meaning

Descriptions are partially synthetic.

Role descriptions appear generated independently from titles.

Trustworthiness

Component	Trust Score
Job Descriptions	MEDIUM

Impact On Ranking

Descriptions should not be treated as absolute truth.

Role consistency checks become important.

Section 2.3 — Skill Bundle Discovery

Objective

Understand skill generation logic.

Methodology

To verify whether skills are the most decisive and realistic component of the dataset, we performed a detailed forensic analysis of the skill field. The process included:

- Extracting and normalizing all skill entries
- Measuring skill frequency distributions
- Identifying common co-occurring skill combinations
- Performing association rule mining to detect structured skill ecosystems
- Comparing skills against candidate titles, summaries, and work descriptions
- Checking for unrealistic or contradictory skill combinations
- Measuring skill consistency across candidate profiles
- Evaluating whether advanced skills appear alongside prerequisite technologies
- Identifying domain-specific skill clusters (AI, Data Engineering, Backend, Frontend, Marketing, Operations, etc.)

The objective was to determine whether skills were randomly assigned or generated according to realistic professional profiles. The analysis focused on validating the authenticity of skill combinations and assessing whether skills could serve as the primary signal for candidate ranking.

Association Rule Mining

Frequent Itemset Discovery

Support Analysis

Evidence

Evidence File:

```
evidence/frequent_skill_bundles.csv
```

Evidence Screenshot:

```
[UPLOAD]
```

Findings

Total Unique Skills

```
133
```

Frequent Skill Bundles

[FILL TOP BUNDLES]

Examples:

- LangChain + RAG
- Pinecone + FAISS
- Spark + Airflow
- React + TypeScript

Meaning

Skills are not randomly assigned.

The generator creates coherent skill ecosystems.

Trustworthiness

Component	Trust Score
Skills	HIGH

Impact On Ranking

Skills should be a primary ranking signal.

Section 2.4 — Archetype Discovery

Objective

Identify hidden candidate categories.

Methodology

[FILL]

Clustering Approach:

[FILL]

Sample Size:

[FILL]

Evidence

Evidence File:

```
evidence/archetypes.csv
```

Evidence Visualization:

```
[UPLOAD]
```

Findings

Number of Archetypes

[FILL]

Major Archetypes

[FILL]

Examples:

- AI Engineer
- Data Engineer
- Backend Engineer
- Marketing Professional
- Operations Manager

Meaning

The dataset is generated around latent candidate personas.

Impact On Ranking

Candidate-to-job matching should operate at archetype level, not merely keyword level.

Section 2.5 — Company Generator Analysis

Objective

Understand company assignment logic.

Methodology

Analyze company frequency distributions, identify recurring naming patterns, and compare generated companies with real-world branding conventions. This includes examining common prefixes, suffixes, industry-related keywords, and organizational naming structures to determine whether companies are synthetic, inspired by real brands, or follow recognizable corporate branding patterns. Additionally, investigate relationships between company names, candidate roles, skills, and industries to uncover generator design logic.

Evidence

```
[UPLOAD]
```

Findings

Companies appear to be assigned from a limited pool of approximately 100 company names that repeat throughout the dataset. The pool contains a mix of real and synthetic company names, with no strong evidence that company assignment is consistently linked to candidate skills, experience, or quality. Due to this repetitive and partially artificial generation process, company names are not a reliable indicator of candidate suitability and should be treated as a weak ranking signal.

Examples:

- Company ↔ Skill relationships
- Industry ↔ Role relationships
- Company Size ↔ Salary relationships

Meaning

[FILL]

Impact On Ranking

[FILL]

Section 2.6 — Role Consistency Analysis

Status

PARTIALLY COMPLETE

Embedding-based analysis pending.

Objective

Measure consistency between:

- Current Title
- Skills
- Summary
- Work Descriptions

Evidence

```
[UPLOAD]
```

Planned Metrics

Title ↔ Description Similarity

[FILL]

Title ↔ Skills Similarity

[FILL]

Title ↔ Summary Similarity

[FILL]

Expected Outcome

Role Consistency Score

Potential ranking feature.

Section 2.7 — Education Generator Analysis

Objective

Understand educational profile generation.

Methodology

[FILL]

Evidence

```
[UPLOAD]
```

Findings

[FILL]

Examples:

- Degree distributions
- Tier distributions
- Education ↔ Skills relationships

Meaning

[FILL]

Impact On Ranking

[FILL]

Section 2.8 — Behavioral Signal Analysis

Objective

Understand behavioral and recruiter-side signals.

Methodology

Signal Distribution Analysis

Missing Value Analysis

Correlation Analysis

Latent Variable Discovery

Findings

Total Signal Columns

24

Missing Value Strategy

Actual Null Values

0

Hidden Missing Values

Github Activity:

```
64,637
```

Offer Acceptance:

```
59,554
```

represented using:

```
-1
```

Meaning

The dataset encodes missing values using sentinel values.

Discovery: Recruiter Interest Dimension

Evidence:

Search Appearance ↔ Saved By Recruiters

```
Correlation ≈ 0.31
```

Signals

- Search Appearance
- Saved By Recruiters
- Profile Views

appear to measure recruiter demand.

Discovery: Candidate Intent Dimension

Evidence:

Open-To-Work candidates show:

- Higher profile views
- Higher recruiter saves
- Higher search appearances

Discovery: Candidate Reliability Dimension

Signals:

- Interview Completion Rate
- Offer Acceptance Rate
- Response Time

Discovery: Verification Effect

Verified candidates show slightly better recruiter engagement metrics.

Discovery: LinkedIn Effect

LinkedIn-connected candidates show:

- More profile views
- More recruiter saves
- Higher search appearances

Discovery: Hidden Candidate Quality Variable

Evidence:

Top candidates consistently score highly across:

- Search Appearance
- Recruiter Saves
- Profile Views
- Interview Completion

This suggests a latent candidate quality factor exists within the generator.

Trust Matrix

Component	Trust
Skills	HIGH
Behavioral Signals	HIGH
Recruiter Signals	HIGH
Career Metadata	MEDIUM
Descriptions	MEDIUM
Education	MEDIUM
Summaries	LOW

Key Dataset Discoveries

Discovery #1

Summaries are heavily templated.

Discovery #2

Descriptions are partially synthetic.

Discovery #3

Skills are coherent and archetype-driven.

Discovery #4

Behavioral signals contain strong ranking information.

Discovery #5

Missing values are encoded using -1.

Discovery #6

A latent candidate quality variable likely exists.

Recommended Features For Ranking System

High Priority

- Skill Match Score
- Recruiter Interest Score
- Candidate Intent Score
- Candidate Reliability Score
- Role Consistency Score
- Archetype Match Score

Medium Priority

- Company Relevance Score
- Education Relevance Score

Low Priority

- Summary Similarity
- Raw Title Match

Remaining Work

Pending

Role Consistency Analysis

Required:

- Title ↔ Skill Similarity
- Title ↔ Description Similarity
- Role Consistency Score

Evidence Required:

```
[UPLOAD]
```

Final Conclusion

The dataset is not random.

It is generated using:

- Reusable text templates
- Skill archetypes
- Behavioral signal models
- Recruiter engagement models
- Latent candidate quality variables

The strongest ranking signals are likely to come from:

1. Skills
2. Behavioral Signals
3. Recruiter Interest Signals
4. Role Consistency
5. Archetype Matching

rather than pure keyword matching or raw embedding similarity.